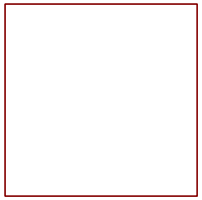
Microcontroller



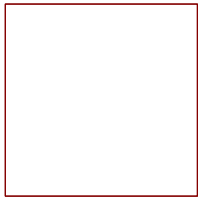
File: Microcontroller.kicad_sch

AnalogFrontend



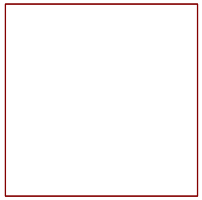
File: AnalogFrontend.kicad_sch

Power



File: Power.kicad_sch

LEDs



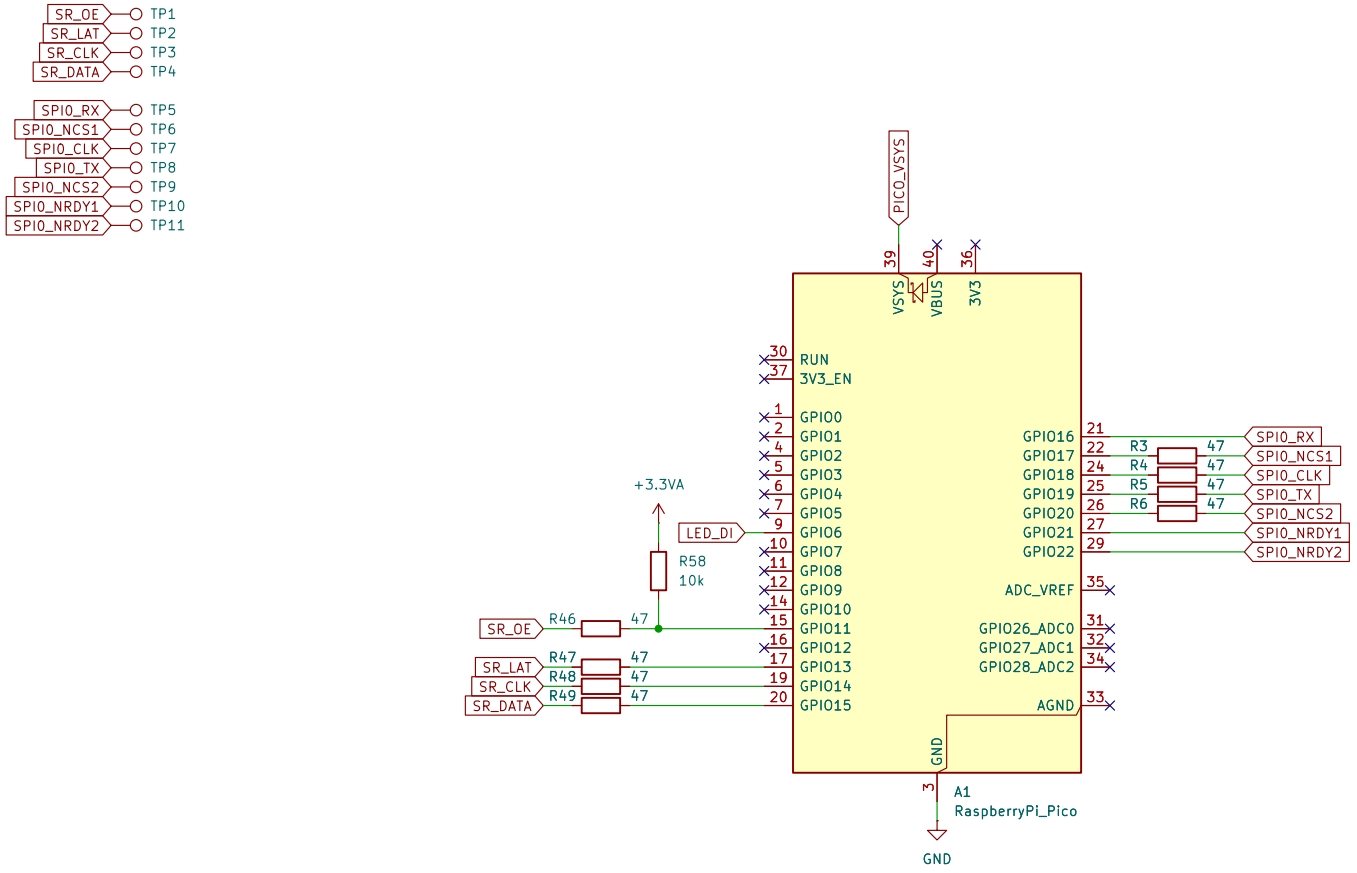
File: leds.kicad_sch

Amon Benson, TU Berlin

Sheet: /
File: ExpressionController.kicad_sch

Title: Expression Adapter

Size: A4	Date: 2026-09-27	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 1/9

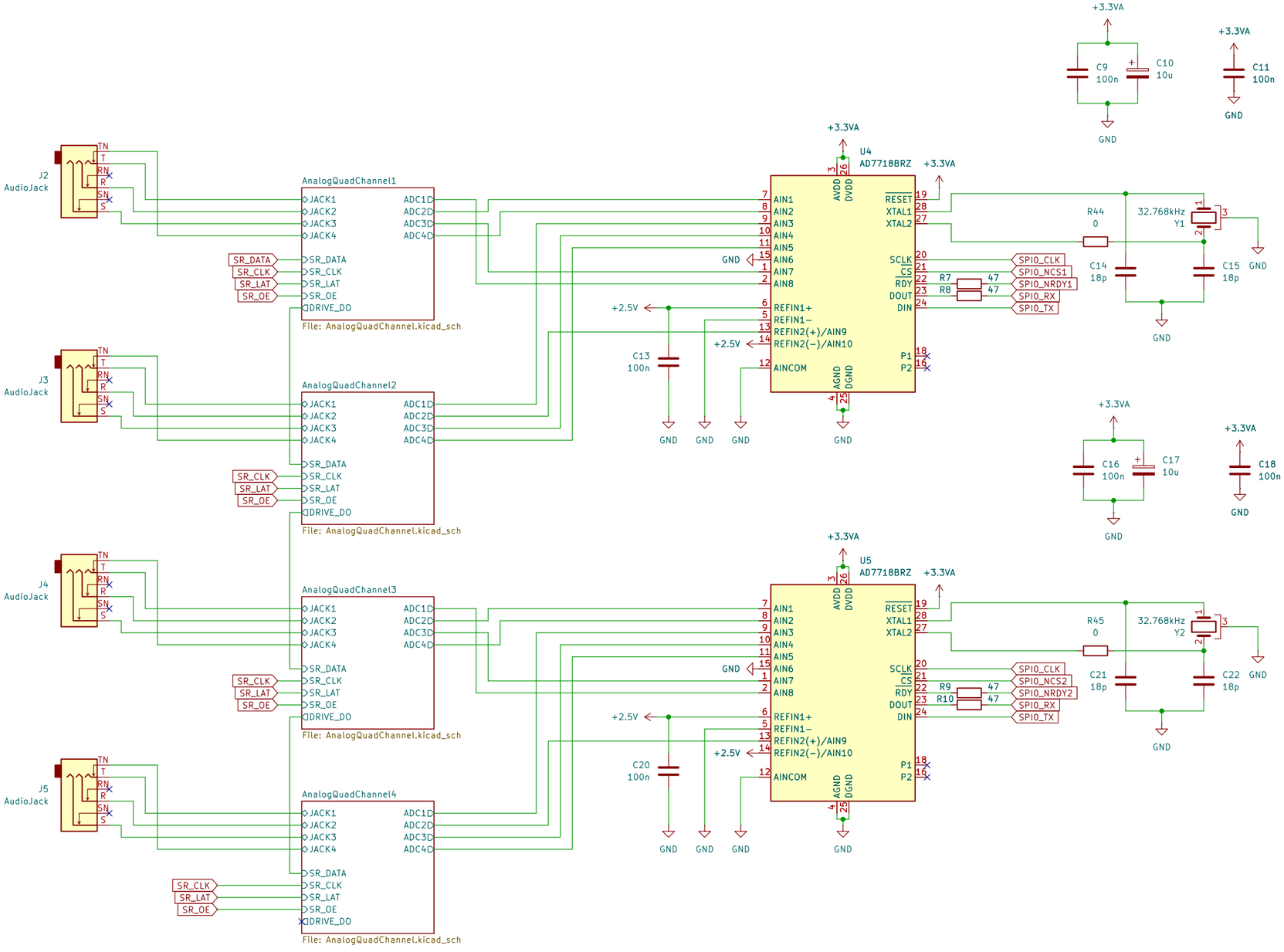


Amon Benson, TU Berlin

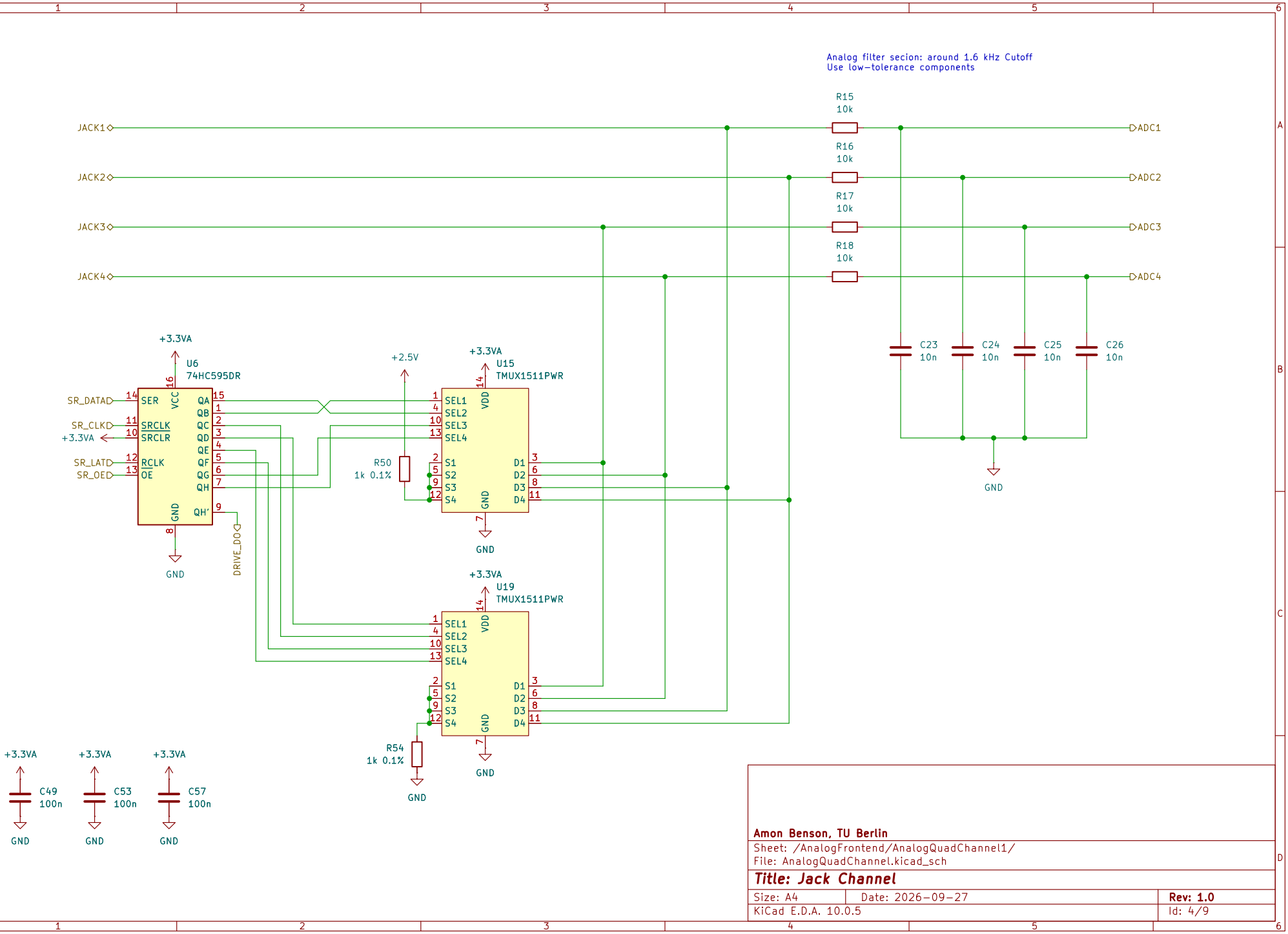
Sheet: /Microcontroller/
File: Microcontroller.kicad_sch

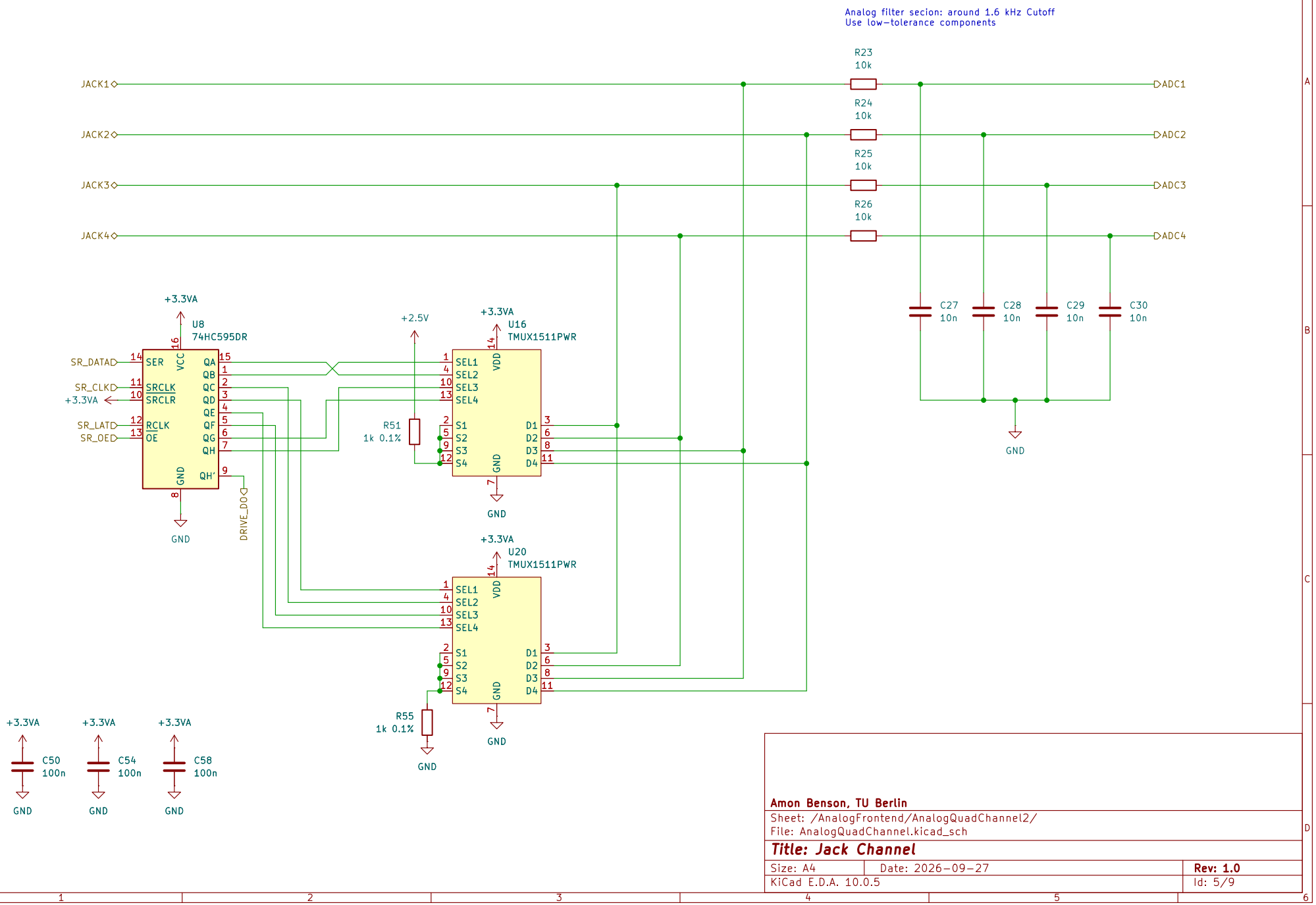
Title: Microcontroller

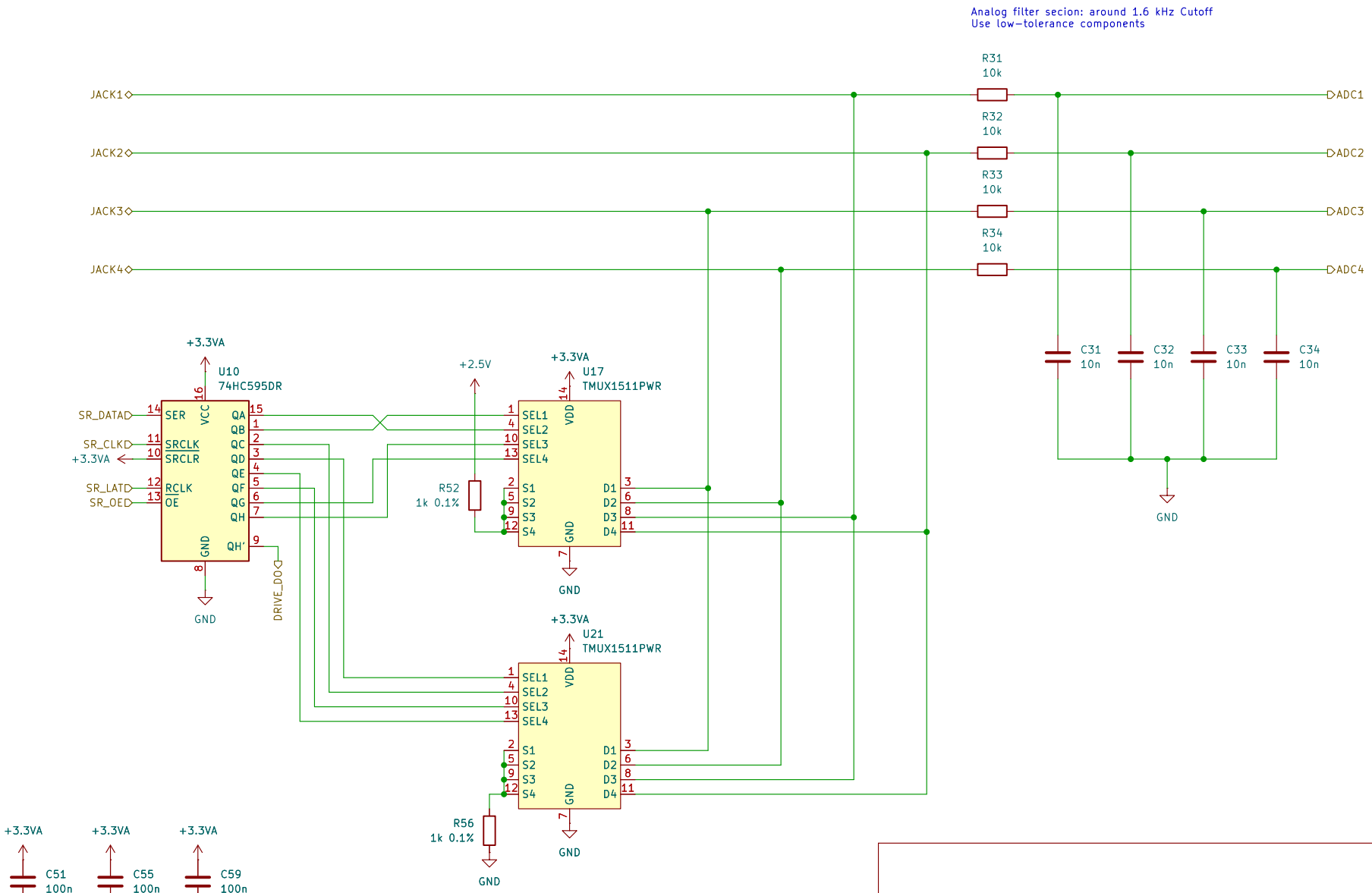
Size: A4	Date: 2026-09-27	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 2/9



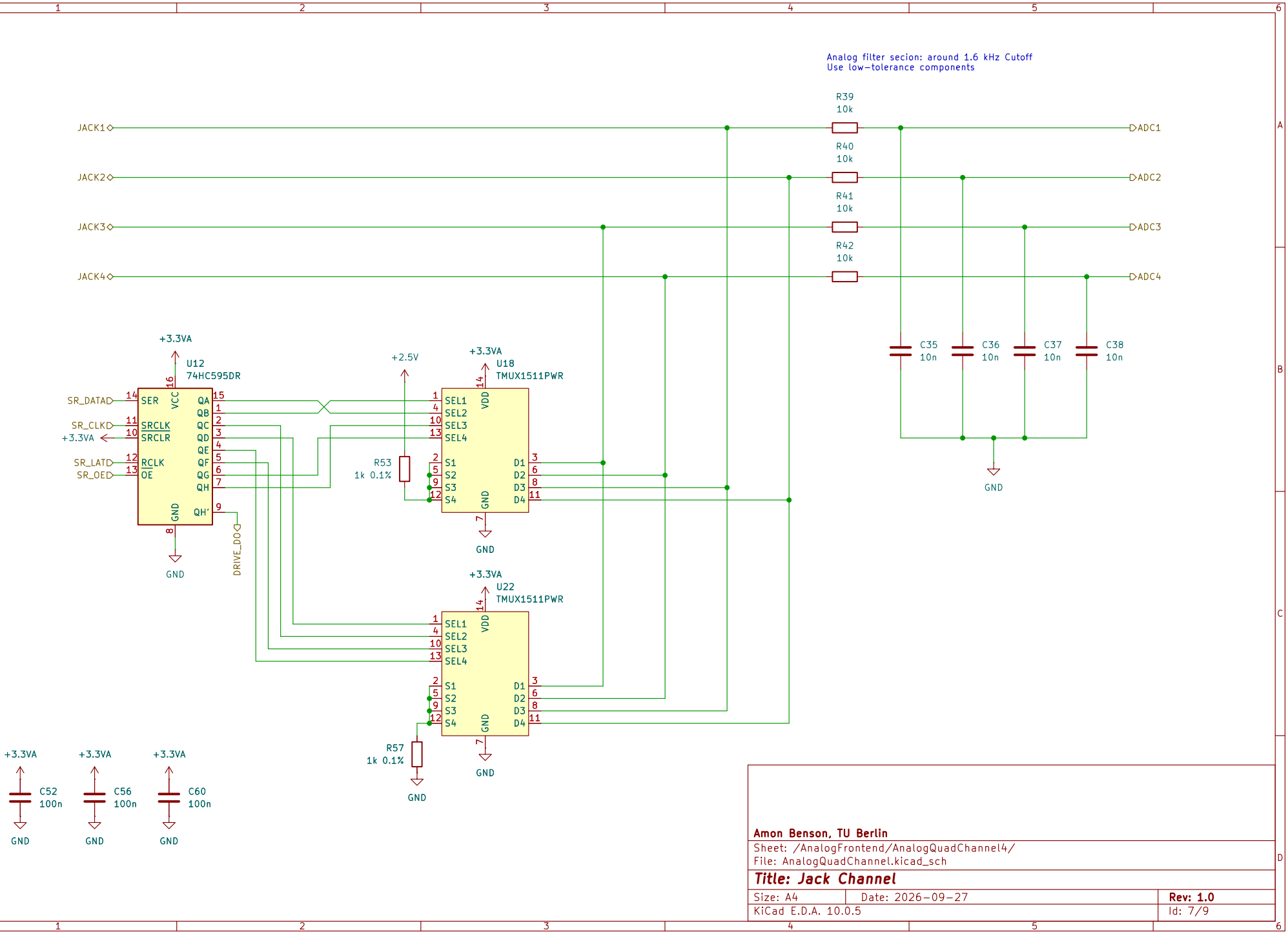
Amon Benson, TU Berlin		
Sheet: /AnalogFrontend/		
File: AnalogFrontend.kicad_sch		
Title: Analog Front End		
Size: A3	Date: 2026-09-27	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 3/9

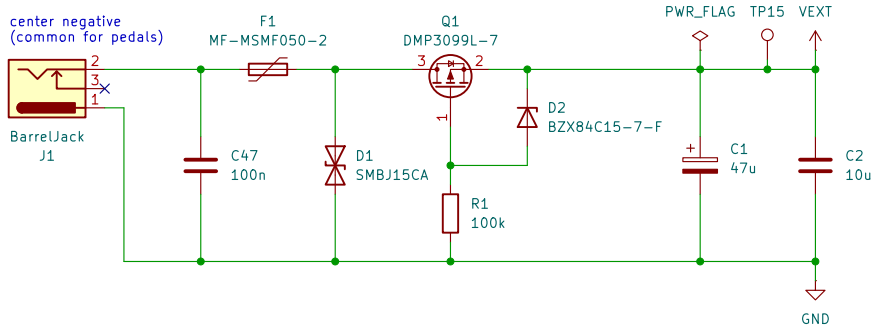

Analog filter secion: around 1.6 kHz Cutoff
Use low-tolerance components
JACK1
JACK2
JACK3
JACK4
R15
10k
R16
10k
R17
10k
R18
10k
ADC1
ADC2
ADC3
ADC4
C23
10n
C24
10n
C25
10n
C26
10n
GND
+3.3VA
U6
74HC595DR
SR_DATA
SR_CLK
SR_LAT
SR_OE
SER
SRCLK
SRCLR
RCLK
OE
VCC
GND
QA
QB
QC
QD
QE
QF
QG
QH
QH'
DRIVE_DO
+2.5V
R50
1k 0.1%
U15
TMUX1511PWR
U19
TMUX1511PWR
SEL1
SEL2
SEL3
SEL4
S1
S2
S3
S4
D1
D2
D3
D4
VDD
R54
1k 0.1%
C49
100n
C53
100n
C57
100n
Amon Benson, TU Berlin
Sheet: /AnalogFrontend/AnalogQuadChannel1/
File: AnalogQuadChannel.kicad_sch
Title: Jack Channel
Size: A4
Date: 2026-09-27
Rev: 1.0
KiCad E.D.A. 10.0.5
Id: 4/9

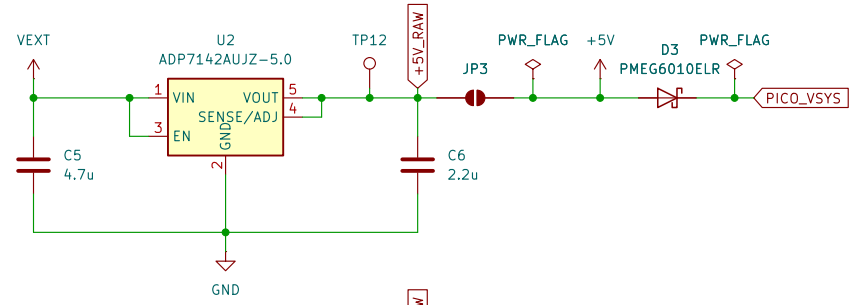

Analog filter secion: around 1.6 kHz Cutoff
Use low-tolerance components
JACK1
JACK2
JACK3
JACK4
R23
10k
R24
10k
R25
10k
R26
10k
ADC1
ADC2
ADC3
ADC4
C27
10n
C28
10n
C29
10n
C30
10n
GND
+3.3VA
U8
74HC595DR
SR_DATA
SR_CLK
SR_LAT
SR_OE
DRIVE_DO
+2.5V
R51
1k 0.1%
U16
TMUX1511PWR
U20
TMUX1511PWR
R55
1k 0.1%
C50
100n
C54
100n
C58
100n
Amon Benson, TU Berlin
Sheet: /AnalogFrontend/AnalogQuadChannel2/
File: AnalogQuadChannel.kicad_sch
Title: Jack Channel
Size: A4
Date: 2026-09-27
Rev: 1.0
KiCad E.D.A. 10.0.5
Id: 5/9

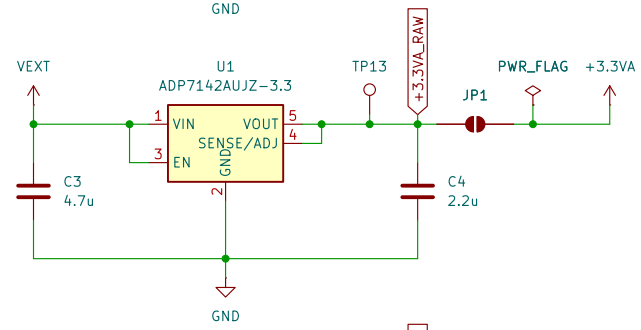


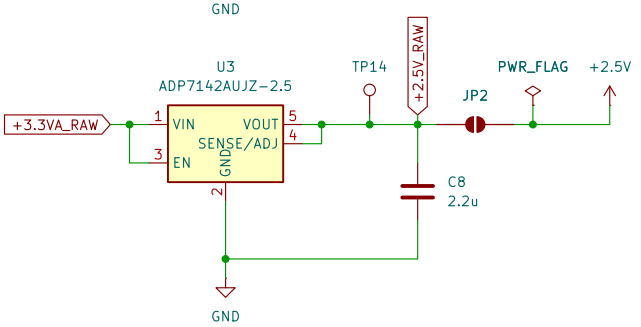
Amon Benson, TU Berlin	
Sheet: /AnalogFrontend/AnalogQuadChannel3/	
File: AnalogQuadChannel.kicad_sch	
Title: Jack Channel	
Size: A4	Date: 2026-09-27
KiCad E.D.A. 10.0.5	Id: 6/9
	Rev: 1.0


Analog filter secion: around 1.6 kHz Cutoff
Use low-tolerance components
JACK1
JACK2
JACK3
JACK4
R39
10k
R40
10k
R41
10k
R42
10k
ADC1
ADC2
ADC3
ADC4
C35
10n
C36
10n
C37
10n
C38
10n
GND
+3.3VA
U12
74HC595DR
SR_DATA
SR_CLK
SR_LAT
SR_OE
DRIVE_DO
+2.5V
R53
1k 0.1%
U18
TMUX1511PWR
U22
TMUX1511PWR
R57
1k 0.1%
C52
100n
C56
100n
C60
100n
Amon Benson, TU Berlin
Sheet: /AnalogFrontend/AnalogQuadChannel4/
File: AnalogQuadChannel.kicad_sch
Title: Jack Channel
Size: A4
Date: 2026-09-27
Rev: 1.0
KiCad E.D.A. 10.0.5
Id: 7/9







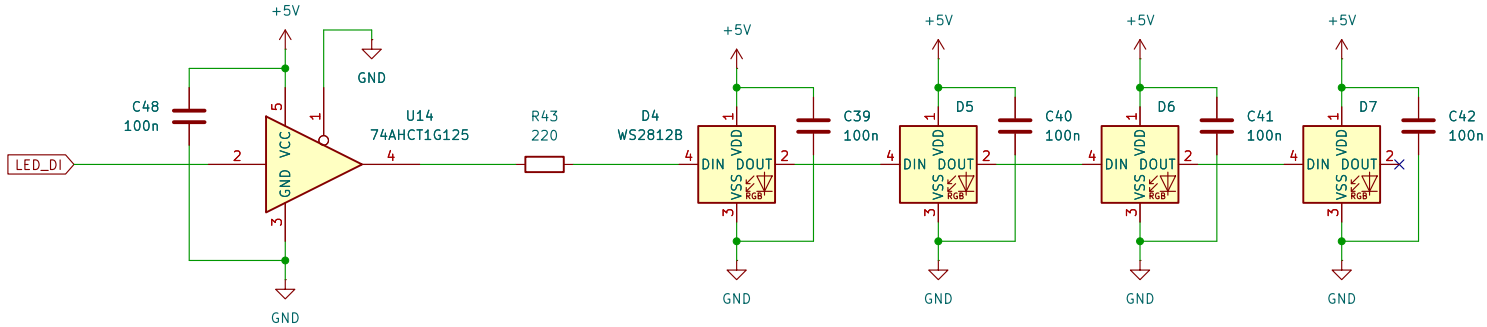


Amon Benson, TU Berlin

Sheet: /Power/
File: Power.kicad_sch

Title: Power

Size: A4	Date: 2026-09-27	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 8/9



Amon Benson, TU Berlin

Sheet: /LEDs/
File: leds.kicad_sch

Title: LEDs

Size: A4	Date: 2026-09-27	**Rev: 1.0**
KiCad E.D.A. 10.0.5		Id: 9/9